

NAY THUTA

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SUMMARY

Driven Engineering with Business graduate with hands-on experience in **systems engineering, embedded development, and cross-functional project coordination**. Skilled in **analyzing complex technical systems**, supporting hardware-software integration, and **managing end-to-end testing and field operations**. Strong background in **data-driven problem solving** and customer engagement, developed through engineering projects and high-volume retail sales environments. **Adept at translating technical concepts for diverse stakeholders** and contributing across **engineering, operations, and customer-facing functions**. **S-Pass eligible and tuition grant bond-compliant** with Singapore-registered employers.

EDUCATION

Diploma in Engineering with Business | Singapore Polytechnic | 3.79

Singapore | 2025

- Developed a versatile skillset, **integrating core engineering principles and foundational business acumen** to solve technical and business challenges.
- Acquired a **Certificate in Entrepreneurship** by cultivating practical skills in **startup design, business model validation, and financial decision-making**.

SKILLS

Electrical Engineering: Circuit Analysis, Embedded Systems, Microcontrollers, System Design

Mechanical Engineering: Mechanics, Thermodynamics, Materials Science

Industrial Engineering: Process Optimization, Lean Manufacturing, ERP Systems

Engineering Software: AutoCAD, SolidEdge, NI Multisim

Programming Languages: C, C++, Java, Python

Data Analytics: Machine Learning, Data Storytelling, KNIME, Power BI, Jupyter Notebook

Business Operations: Project Management, Financial Reporting, Microsoft Office Suite

Sales & Marketing: Digital Marketing, Market Analysis, Solution Selling, Technical Presentations

Languages: English (Fluent), Burmese (Fluent)

EXPERIENCE

Project Intern | Singapore Polytechnic | Dover | March 2024 – August 2024

- Coordinated a **22-week autonomous drone delivery project**, collaborating cross-functionally with stakeholders to ensure **on-schedule development and successful execution of project milestones**.
- Designed and implemented the custom SPFly plugin by **analyzing 10k+ lines of open-source code**, enabling mobile app communication and enhancing system usability.
- Managed **50+ hours of on-site flight test operations** to validate system functionality and safety compliance, **resulting in 100% incident-free record**.
- Presented system capabilities and project outcomes to **technical audiences** via **high-stakes presentations and a live proof-of-concept demo**.

Part-time Coffee Specialist | Nespresso | Islandwide | December 2022 - March 2025

- Delivered **consultative sales support to 1000+ customers**, consistently exceeding sales expectations by **rapidly adapting to changing customer profiles** with a client-first attitude in high-traffic retail environments.
- Communicated **Nespresso's technology and sustainability values** by conducting **product demos** and executing **marketing campaigns** to strengthen client relationships and brand presence.

Part-time Service Crew | Fairmont Singapore & Swissôtel The Stamford | Downtown Core | May 2022 - March 2025

- Sustained **exemplary professionalism and operational efficiency** in fast-paced, high-pressure luxury dining environments, successfully upholding service quality throughout **demanding 12-hour shifts**.
- Fostered a positive service atmosphere by **cultivating strong rapport with customers and colleagues**, directly contributing to enhanced team performance and guest satisfaction.

PROJECT

Sustainable Irrigation System – Microcontrollers, Embedded C, AutoCAD, Circuit Design

| naythuta.github.io/my-portfolio/projects/sustainable-irrigation-system/

- Engineered a **PIC18F4550-based smart irrigation system** by integrating hardware components through **hands-on circuit analysis and design** to implement robust sensor interfacing and actuator drive mechanisms, optimizing water usage and enhancing climate resilience.
- Designed an **M&E system prototype** by creating precise system layouts in **AutoCAD** to map the microcontroller and 8 key I/O components; executed the physical assembly via detailed **stripboard planning and manual soldering** to ensure seamless integration.

Wormbrews: Sustainable Fertilizer Startup - Entrepreneurship, Prototyping, Market Research

| naythuta.github.io/my-portfolio/projects/wormbrews/

- Co-developed a **vermicomposting product concept** emphasizing compactness, odor control, and affordability to address urban living constraints and support food waste reduction.
- Led **qualitative and quantitative market research with 20+ participants**, utilizing a prototype to **identify key pain points and validate 86% interest** in eco-friendly composting among urban gardeners.

Pharmaceutical Sentiment Analysis - Machine Learning, Python, Jupyter Notebook, Microsoft Power BI

| naythuta.github.io/my-portfolio/projects/patient-sentiment-analysis/

- Built a **sentiment analysis pipeline on 27k+ drug reviews** using VADER and KMeans to uncover patterns between patient sentiment and satisfaction ratings.
- Delivered **actionable insights via Power BI dashboards**, helping pharmaceutical companies track drug perception and support patient-centric decisions.

Habit Heroes: Gamified Habit Tracker – Mobile Application Development, Java, Android Studio

- Engineered a **native Android application in Java** with 10+ advanced features to promote general well-being by facilitating **gamified habit tracking and parental monitoring** through a hybrid backend using **Firestore Realtime Database and SQLite**.
- Implemented complex system integrations, including the **OpenAI API** for personalized AI wellness advice, **Google Maps SDK** for safety tracking, **Biometric Security** for privacy, and **custom background services** to monitor screen time usage across external applications.

Data Storytelling for Strategic Drone Site Selection – Microsoft Power BI, Geospatial Analysis

- Designed and deployed an **interactive Power BI dashboard**, analyzing survey data from 200 residents to model the correlation between food insecurity risks and physical inaccessibility within the Age 50+ demographic.
- Presented a **compelling, data-driven narrative**, recommending two pilot sites based on the highest average distance to amenities to **maximize the social impact** for the vulnerable population.

DARTSDelivery: Drone-Assisted Last-Mile Delivery - Entrepreneurship, Market Research, Web Prototype

- Co-developed the business concept for a **drone-based last-mile delivery service** by analyzing validation data from 100+ survey respondents, **refining value propositions, and shaping the customer journey** around the 88% who reported missed parcel issues.
- Designed **the entire website prototype on Wix.com**, translating the concept into a functional multi-page platform to **visualize service workflow, B2B onboarding, and operational requirements**, supporting pitch presentations and improving stakeholder clarity of the solution.

Digital Marketing Plan for CocoNutnut - Market Analysis, Customer Segmentation, Content Strategy

- Spearheaded market strategy through **data-driven market analysis (STEER, SWOT, 4Ps)** for an F&B brand, defining a core target segment and establishing **the Online Value Proposition** to strategically capture growth in the health & wellness sector.
- Co-developed a **comprehensive 2-week Editorial Calendar** detailing 9 distinct content pieces across 3 content pillars for 4 online platforms (including paid and organic campaigns); identified **critical brand weaknesses** to recommend targeted operational and sustainable supply chain improvements.

Digital Accounting System Implementation - Accounting, Financial Reporting, ABSS Premier

- Executed the digitization of a company's accounting functions using **ABSS Premier**, setting up the Chart of Accounts and translating a full month of source documents into general journal entries to **migrate the business from manual to digital operations**.
 - Generated and validated **key financial statements** (Trial Balance, Statement of Profit or Loss, and Statement of Financial Position) with **zero errors**, reporting a Gross Profit of S\$53,500.00 for the first period of computerization.
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